12TH CHEMISTRY · CHAPTER 25 · SOLVED EXERCISE

Organic Synthesis

Complete textbook exercise (MCQs, Short Answer, Constructed Response, Descriptive) with Quick Check solutions

Q1. MULTIPLE CHOICE QUESTIONS:

I) After disconnection the components obtained are called:

(a) target molecule (b) synthons
(c) synthetic equivalents (d) starting material

Answer: (b) synthons — Synthons are the fragments of the target molecule generated by a disconnection.

II) How will you convert the amide functional group (-C≡N) to the amine group?

(a) by an oxidation reaction (b) by a reduction reaction
(c) by an acid hydrolysis (d) by an addition reaction

Answer: (b) by a reduction reaction — Reduction with $LiAlH_4$ gives the amine.

The stem calls –C≡N the "amide functional group"; –C≡N is the nitrile group (the amide group is –$CONH_2$). Both are converted into amines by reduction, so the answer is the same.

III) What does this arrow (⟹) mean in organic chemistry reactions?

(a) synthesis (b) Retrosynthesis
(c) reversible reaction (d) disconnection

Answer: (b) Retrosynthesis — The open-ended arrow is the retrosynthetic arrow.

IV) Which reagent you will choose to convert aldehyde group into alcohol?

(a) $KMnO_4/H^+$
(b) $NaBH_4/H_3O^+$
(c) Zn/HCl
(d) Na/Liquid NH_3

Answer: (b) $NaBH_4/H_3O^+$ — $NaBH_4$ reduces aldehydes to primary alcohols.

V) Which reagent is commonly used to oxidize primary alcohol into carboxylic acid?

(a) H_2, Pd/C
(b) $LiAlH_4$
(c) $K_2Cr_2O_7$, H_2SO_4
(d) $NaBH_4$

Answer: (c) $K_2Cr_2O_7$, H_2SO_4 — Acidified potassium dichromate is the oxidising agent; the others are reducing agents.

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VI) When toluene is treated with $KMnO_4/H_2SO_4$, we obtain:

(a) benzyl chloride (b) benzene sulphonic acid
(c) benzoic acid (d) acetophenone

Answer: (c) benzoic acid — The methyl side chain is oxidised to –COOH.

VII) Which functional group is formed when a carboxylic acid reacts with an alcohol in the presence of an acid as catalyst?

(a) alkane (b) alcohol
(c) aldehdye (d) ester

Answer: (d) ester — This is esterification.

VIII) What is compound A in the given sequence?
phenol → (nitration) nitrophenol → (reduction) aminophenol → (acetylation) A

(a) Aspirin (b) Paracetamol
(c) Acetophenone (d) Halicin

Answer: (b) Paracetamol — Acetylation of 4-aminophenol with acyl chloride gives paracetamol.

IX) In organic synthesis, what is the primary purpose of a 'Functional Group Interconversion' (FGI)?

(a) To break the molecule into smaller fragments
(b) To increase total number of carbon atoms in molecule
(c) To make the molecule more reactive
(d) To change the functional group into another to enable a specific reaction

Answer: (d) To change the functional group into another to enable a specific reaction — FGI is the conversion of one functional group into another to reach the target molecule.

X) The drug 'Halicin' developed by AI work as an:

(a) antiviral drung (b) antifungal drug
(c) antibiotic drug (d) anticancer drug

Answer: (c) antibiotic drug — It killed many disease-causing bacteria that were resistant to treatment.

Q2. SHORT ANSWER QUESTIONS

a) Differentiate between synthesis and retrosynthesis.

Synthesis	Retrosynthesis
The target molecule is prepared starting from simple, commercially available compounds.	The target molecule is broken down (on paper) into simpler, readily available compounds.
It moves forward: starting material → target molecule.	It moves backward: target molecule ⟹ starting material.
It consists of actual reactions with reagents and conditions.	It consists of imaginary disconnections and functional group interconversions.
It is shown by the ordinary reaction arrow (→).	It is shown by the open-ended retrosynthetic arrow (⟹).

b) How a carboxylic group is converted to an amide group?

The carboxylic acid is first converted into its acid halide with $SOCl_2$, and the acid halide is then treated with ammonia.

$$CH_3\text{–}COOH + SOCl_2 \rightarrow CH_3\text{–}COCl + SO_2 + HCl$$
$$CH_3\text{–}COCl + NH_3 \rightarrow CH_3\text{–}CO\text{–}NH_2 + HCl$$

c) How synthon and synthetic equivalent are related to each other?

A synthon is a fragment of the target molecule generated by a disconnection; it is often an ionic species and is usually too unstable to be used itself. A synthetic equivalent is the real reagent that carries out the function of that synthon. For example, the synthon CH_3CO^+ has acyl chloride, CH_3COCl, as its synthetic equivalent, and the synthon NO_2^+ has nitric acid.

d) Write down the formula of the reagent used for oxidation of organic compounds.

Acidified potassium dichromate, $\mathbf{K_2Cr_2O_7}$ / $\mathbf{H_2SO_4}$. Acidified potassium manganate(VII), $\mathbf{KMnO_4}$ / $\mathbf{H_2SO_4}$, is also used (e.g. for toluene → benzoic acid).

e) Show with equations how alcohol can be converted into amines, alkenes and halogenoalkanes.

Halogenoalkane:

$$3CH_3\text{–}CH_2\text{–}OH + PBr_3 \rightarrow 3CH_3\text{–}CH_2\text{–}Br + H_3PO_3$$

Amine (through the halogenoalkane):

$$CH_3\text{–}CH_2\text{–}Br + NH_3 \rightarrow CH_3\text{–}CH_2\text{–}NH_2 + HBr$$

Alkene:

$$CH_3\text{–}CH_2\text{–}OH \rightarrow CH_2{=}CH_2 + H_2O \text{ (conc. } H_2SO_4\text{, heat)}$$

f) What is a disconnection in retrosynthesis of a molecule?

Disconnection is a process which involves the imaginary breaking down of the target molecule progressively into simpler starting materials. Those bonds are disconnected which give a considerable structural simplification; the fragments obtained are called synthons.

g) Mention one useful application of FGI in organic synthesis?

FGI is used to bring in a group that cannot be attached directly. For example, in the synthesis of aniline the amino group is obtained from the nitro group: it is easy to attach $–NO_2$ to benzene by electrophilic substitution, and it is then reduced to $–NH_2$.

h) Give one method for the reduction of a nitro group ($-NO_2$) to an amino group ($-NH_2$).

The nitro compound is reduced with tin and hydrochloric acid (Sn + HCl).

$$C_6H_5–NO_2 + 6[H] \rightarrow C_6H_5–NH_2 + 2H_2O \ (Sn + HCl)$$

i) How a carboxylic group is converted to an alcohol?

By reduction with lithium aluminium hydride, $LiAlH_4$; a primary alcohol is formed.

$$CH_3–COOH + 4[H] \rightarrow CH_3–CH_2–OH + H_2O \ (LiAlH_4)$$

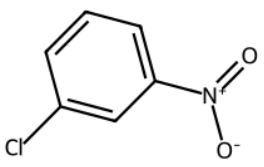
j) Design synthesis of 3-chloronitrobenzene and 2-chloronitrobenzene starting from benzene in two steps.

3-Chloronitrobenzene: nitrate first, then chlorinate. The $–NO_2$ group is meta directing, so chlorine enters position 3.

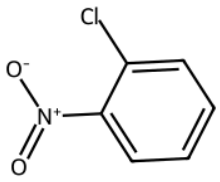
$$C_6H_6 + HNO_3 \rightarrow C_6H_5–NO_2 + H_2O \ (H_2SO_4)$$
$$C_6H_5–NO_2 + Cl_2 \rightarrow 3\text{-}Cl–C_6H_4–NO_2 + HCl \ (FeCl_3)$$

2-Chloronitrobenzene: chlorinate first, then nitrate. The –Cl atom is ortho-para directing, so the nitro group enters position 2 (the 4-isomer formed with it is separated).

$$C_6H_6 + Cl_2 \rightarrow C_6H_5–Cl + HCl \ (FeCl_3)$$
$$C_6H_5–Cl + HNO_3 \rightarrow 2\text{-}Cl–C_6H_4–NO_2 + H_2O \ (H_2SO_4)$$



3–chloronitrobenzene

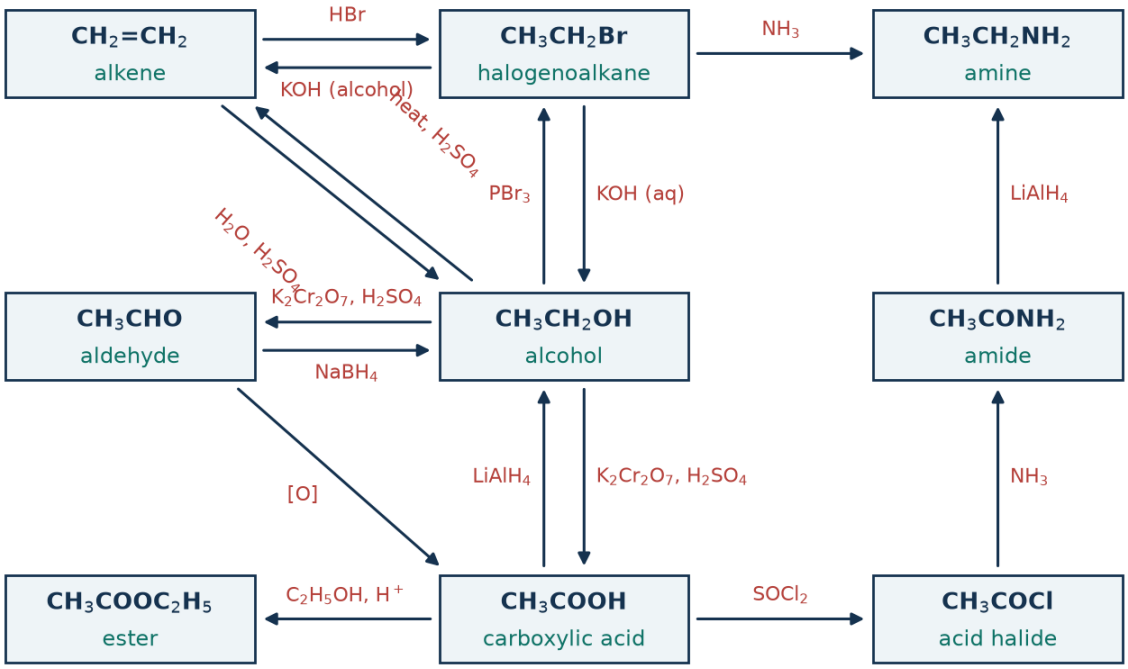


2–chloronitrobenzene

Q3. CONSTRUCTED RESPONSE QUESTIONS

a) Construct the functional group interconversion flow sheet and highlight the important reagents and conversions involved.

The flow sheet below shows how the main aliphatic functional groups are converted into one another; the reagent is written on each arrow.



Functional group interconversion in aliphatic organic compounds

Conversion	Reagent(s)
Alkene → halogenoalkane	HBr
Halogenoalkane → alkene	KOH (alcoholic)
Alkene → alcohol	H_2O, H_2SO_4
Alcohol → alkene	heat, H_2SO_4
Halogenoalkane → alcohol	KOH (aq)
Alcohol → halogenoalkane	PBr_3
Halogenoalkane → amine	NH_3
Alcohol → aldehyde → carboxylic acid	$K_2Cr_2O_7$, H_2SO_4
Aldehyde → alcohol	$NaBH_4$
Carboxylic acid → alcohol	$LiAlH_4$
Carboxylic acid → ester	C_2H_5OH, H^+
Carboxylic acid → acid halide	$SOCl_2$

Conversion	Reagent(s)
Acid halide → amide	NH_3
Amide → amine	$LiAlH_4$

In Figure 25.3 the textbook labels the aldehyde → carboxylic acid arrow "V_2O_5" and also draws NH_3 arrows from the alcohol to the amine and amide; these are not reactions described in the earlier chapters and are not included here.

b) Construct the retrosynthetic tree for the paracetamol and then design its synthesis.

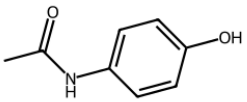
Step I – Disconnection: paracetamol is disconnected at the C–N bond of the amide group to get the synthons.

Step II – Synthetic equivalents: 4-aminophenol and acyl chloride. The bond is broken at this position because the synthetic equivalents are well known to react through a nucleophilic addition–elimination reaction.

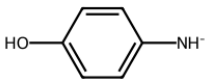
Step III – Functional group interconversion: 4-aminophenol ⟹ 4-nitrophenol, because it is easy to prepare p-nitrophenol, which can then be reduced to p-aminophenol.

Step IV – Disconnection: 4-nitrophenol is disconnected at the C–N bond, creating the 4-hydroxy aryl anion and the nitronium cation (NO_2^+).

Step V – Synthetic equivalents: phenol and nitric acid.

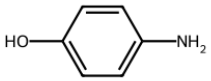


TM: paracetamol



synthon A

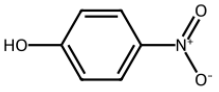
synthon B



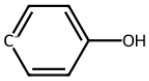
SE of A: 4-aminophenol



SE of B: acyl chloride

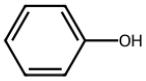


after FGI: 4-nitrophenol



synthon A: 4-hydroxy aryl anion

synthon B: nitronium ion



SE of A: phenol (B: nitric acid)

Synthesis of paracetamol from phenol (starting material):

$$HO–C_6H_5 + HNO_3 \rightarrow HO–C_6H_4–NO_2 + H_2O\ (H_2SO_4)$$

$$HO–C_6H_4–NO_2 + 6[H] \rightarrow HO–C_6H_4–NH_2 + 2H_2O\ (Sn + HCl)$$

$$HO–C_6H_4–NH_2 + CH_3–COCl \rightarrow HO–C_6H_4–NH–CO–CH_3 + HCl$$

DESCRIPTIVE QUESTIONS

Q4. Give a detailed account of importance of retrosynthesis in organic synthesis.

Retrosynthesis is exactly the opposite of synthesis. It involves the breaking down of the target molecule into less complicated, simpler and more readily available compounds which can serve as starting materials. It uses the **disconnection approach**: those bonds of the target molecule are disconnected which result in a considerable structural simplification. Following a series of disconnections, we eventually arrive at simple, easily available starting materials.

Importance:

- It helps to select the best route: a good synthesis converts cheap and easily available starting materials into the product by the least number of steps and with a high overall yield.
- By disconnecting the bonds which give the greatest structural simplification, a complicated target molecule is traced back, step by step, to simple and easily available starting materials.
- It shows which synthons are needed and therefore which real reagents (synthetic equivalents) and reactions must be used in each step.
- It shows where a functional group interconversion is required (e.g. $-NH_2$ from $-NO_2$).
- It tells how many and which steps are actually needed, e.g. to prepare paracetamol or aspirin from phenol.
- Its most important application is in the synthesis of pharmaceutical products: complex medicinal compounds become simple and achievable after a series of efficient retrosynthetic steps.

Terminology used:

Term	Meaning
Retrosynthetic arrow	An open-ended arrow, denoted by $\Longrightarrow$.
Target molecule (TM)	The molecule which is to be prepared by disconnection.
Disconnection	A process which involves the imaginary breaking down of the TM progressively into simpler starting materials.
Synthons	Fragments of the target molecule which are generated by a disconnection. They are often ionic species and represent the building blocks for the synthesis.
Synthetic equivalent (SE)	A reagent used to carry out the function of a synthon, which cannot be used itself because it is usually unstable.
Functional group interconversion (FGI)	The conversion of one functional group into another to reach the target molecule.

Term	Meaning
Retrosynthetic tree	A diagram showing all possible disconnections and synthons from the target molecule to the starting materials.

Q5. Explain the role of halogenoalkanes in retrosynthesis of organic molecules.

Halogenoalkanes are key intermediates in the functional group interconversion grid, because they are easily made from readily available compounds and easily converted into many other functional groups. When a target molecule is disconnected or its functional group is interconverted, a halogenoalkane very often appears as the compound one step back.

They are easily prepared from alkenes (HBr) and from alcohols, which are more easily available starting materials:

$$CH_2{=}CH_2 + HBr \rightarrow CH_3{-}CH_2{-}Br$$
$$R{-}CH_2{-}OH \rightarrow R{-}CH_2{-}Cl\ (SOCl_2)$$

They are converted into other functional groups:

Target functional group	Reagent	Equation
Alcohol	KOH (aq)	$CH_3CH_2Br + KOH \rightarrow CH_3CH_2OH + KBr$
Amine	NH_3	$CH_3CH_2Br + NH_3 \rightarrow CH_3CH_2NH_2 + HBr$
Alkene	KOH (alcohol)	$CH_3CH_2Br + KOH \rightarrow CH_2{=}CH_2 + KBr + H_2O$

They are synthetic equivalents in carbon–carbon bond formation: an alkyl halide acts for an alkyl cation synthon, e.g. CH_3Cl with $AlCl_3$ converts benzene into toluene. Thus the textbook example of a primary amine is traced back as: amine $\Longrightarrow$ haloalkane $\Longrightarrow$ alcohol.

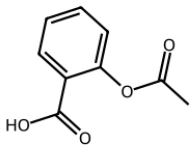
Q6. Construct the retrosynthetic tree for the aspirin and then design its synthesis.

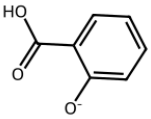
Step I – Disconnection-I: aspirin is disconnected at the C–O bond of the ester linkage to get the synthons.

Step II – Synthetic equivalents: salicylic acid and acetic anhydride.

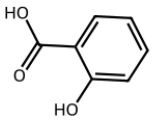
Step III – Disconnection-II: the C–C bond of the carboxylic group of salicylic acid is disconnected.

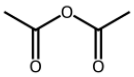
Step IV – Synthetic equivalents: phenol and carbon dioxide.

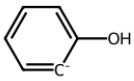

TM: aspirin


synthon A

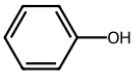
synthon B


SE of A: salicylic acid


SE of B: acetic anhydride


synthon A (from salicylic acid)

synthon B


SE of A: phenol (B: carbon dioxide)

Synthesis of aspirin:

$$C_6H_5\text{–}OH + CO_2 \rightarrow HO\text{–}C_6H_4\text{–}COOH \text{ ((i) high T and P; (ii) } H_2SO_4)$$
$$HO\text{–}C_6H_4\text{–}COOH + (CH_3CO)_2O \rightarrow CH_3COO\text{–}C_6H_4\text{–}COOH + CH_3COOH \ (H_3PO_4)$$

Phenol is first converted into salicylic acid, which is then heated with acetic anhydride and H_3PO_4 to give aspirin (acetic acid is the by-product).

Q7. Design the synthesis of the following molecules starting from benzene. a) $C_6H_5\text{–}NH_2$ b) $C_6H_5\text{–}CH_3$ c) $C_6H_5\text{–}COOH$ d) $C_6H_5\text{–}CH_2\text{–}OH$

a) Aniline: nitration followed by reduction.

$$C_6H_6 + HNO_3 \rightarrow C_6H_5\text{–}NO_2 + H_2O \ (H_2SO_4)$$
$$C_6H_5\text{–}NO_2 + 6[H] \rightarrow C_6H_5\text{–}NH_2 + 2H_2O \ (Sn, HCl)$$

b) Toluene: Friedel–Crafts alkylation.

$$C_6H_6 + CH_3Cl \rightarrow C_6H_5\text{–}CH_3 + HCl \ (AlCl_3)$$

c) Benzoic acid: toluene is prepared as in (b) and its side chain is oxidised.

$$C_6H_5\text{–}CH_3 + 3[O] \rightarrow C_6H_5\text{–}COOH + H_2O \ (KMnO_4, H_2SO_4)$$

d) Benzyl alcohol: toluene is prepared as in (b), chlorinated in the side chain in light and then hydrolysed.

$$C_6H_5\text{–}CH_3 + Cl_2 \rightarrow C_6H_5\text{–}CH_2\text{–}Cl + HCl \ (light)$$
$$C_6H_5\text{–}CH_2\text{–}Cl + KOH_{(aq)} \rightarrow C_6H_5\text{–}CH_2\text{–}OH + KCl$$

Q8. How will you carry out the following conversions?

a) $CH_3—CH_3 \rightarrow CH_3—COOH$

$$CH_3–CH_3 + Cl_2 \rightarrow CH_3–CH_2–Cl + HCl \text{ (light)}$$
$$CH_3–CH_2–Cl + KOH_{(aq)} \rightarrow CH_3–CH_2–OH + KCl$$
$$CH_3–CH_2–OH + 2[O] \rightarrow CH_3–COOH + H_2O \ (K_2Cr_2O_7,\ H_2SO_4)$$

b) $CH_2=CH_2 \rightarrow CH_3—CH_2—OH$

$$CH_2=CH_2 + H_2O \rightarrow CH_3–CH_2–OH \ (H_2SO_4)$$

c) $CH_3—CH_2—CH_2—Cl \rightarrow CH_3—CH=CH_2$

$$CH_3–CH_2–CH_2–Cl + KOH \rightarrow CH_3–CH=CH_2 + KCl + H_2O \text{ (KOH in alcohol, heat)}$$

d) $CH_3—COOH \rightarrow CH_3—CH_2—OH$

$$CH_3–COOH + 4[H] \rightarrow CH_3–CH_2–OH + H_2O \ (LiAlH_4)$$

QUICK CHECK QUESTIONS (IN-CHAPTER)

These questions appear in the "Quick Check" boxes inside the chapter, not in the end-of-chapter exercise.

Quick Check 25.1

i. Why aldehydes and ketones undergo nucleophilic addition reaction readily?

The carbonyl group is polar: the carbon atom carries a partial positive charge (δ+) and is electrophilic, so it is easily attacked by nucleophiles. The π bond of the C=O group is easily broken, and the planar shape of the group leaves the carbonyl carbon uncrowded. Hence addition of nucleophiles takes place readily.

ii. Convert a carboxylic acid to its anhydride.

The acid is first converted into its acid chloride (an FGI), which is then treated with the sodium salt of the same acid:

$$CH_3–COOH + SOCl_2 \rightarrow CH_3–COCl + SO_2 + HCl$$
$$CH_3–COCl + CH_3–COONa \rightarrow (CH_3CO)_2O + NaCl$$

The preparation of acid anhydrides is not described in the textbook; this is the standard method. (Acetic anhydride appears in the book only as a reagent for aspirin.)

Quick Check 25.2

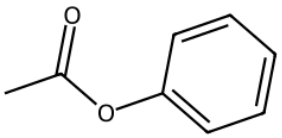
a) Why nitration of benzene is done during synthesis of aniline?

It is easy to attach a nitro group to the benzene ring by electrophilic substitution, whereas an amino group cannot be attached directly by nucleophilic substitution. So benzene is first nitrated, and the nitro group is then reduced (Sn + HCl) to the amino group.

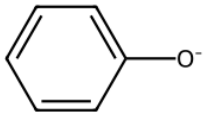
b) Phenyl ethanoate is an important molecule used in synthesis of medicines for hepatitis and various infection. i) Draw its retrosynthetic tree. ii) Write structures of its synthetic equivalents. iii) Design its linear synthesis.

i) Retrosynthetic tree: the C–O bond of the ester linkage is disconnected, giving synthon A (phenoxide ion) and synthon B (acylium ion).

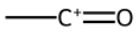
ii) Synthetic equivalents: phenol, C_6H_5–OH, for synthon A and ethanoyl (acyl) chloride, CH_3–COCl, for synthon B.



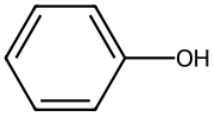
TM: phenyl ethanoate



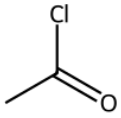
synthon A: phenoxide ion



synthon B: acylium ion



SE of A: phenol



SE of B: ethanoyl chloride

iii) Linear synthesis (from benzene, using the conversions of Figure 25.4):

$$C_6H_6 \rightarrow C_6H_5\text{–}SO_3H\ (SO_3,\ H_2SO_4)$$

$$C_6H_5\text{–}SO_3H \rightarrow C_6H_5\text{–}OH\ ((i)\ NaOH,\ (ii)\ HCl)$$

$$C_6H_5\text{–}OH + CH_3\text{–}COCl \rightarrow CH_3\text{–}COO\text{–}C_6H_5 + HCl$$

Quick Check 25.3

Aspirin and Paracetamol, both are synthesized starting from phenol. I) What is the overall functional group difference in them? ii) In synthesis of which FGI (Functional Group Interconversion) is involved? iii) Which synthon is common in preparation of them? Write down its structure and also mention its most common synthetic equivalent.

I) Aspirin contains an **ester** group and a **carboxylic acid** group on the benzene ring. Paracetamol contains an **amide** group and a phenolic **hydroxyl** group.

ii) In the synthesis of **paracetamol**: the amino group of 4-aminophenol is obtained from the nitro group of 4-nitrophenol (reduction with Sn + HCl).

iii) The acylium (acetyl) cation, CH_3–C^+=O, is the common synthon. Its most common synthetic equivalent is acyl chloride (ethanoyl chloride, CH_3COCl); in the aspirin synthesis acetic anhydride is used.